

Callum M. Scott

07930 333 257
callumscottg@gmail.com
linkedin.com/in/callumscott
callumscott.github.io

ABOUT ME

Self-taught developer with a background in mathematics and physics. Recent work involves evaluating LLM performance on mathematical reasoning tasks. Built an n-body gravity simulation in Python and two full-stack web apps with Node.js, Express, and PostgreSQL, with an ongoing interest in type systems and formal verification. Looking to contribute to a strong team and grow fast as a developer.

PROJECTS

3D N-BODY GRAVITY SIMULATION

3D gravitational simulation built with Python, NumPy, and visualised with Matplotlib. Reducing error accumulation over 1000s of timesteps motivated the use of Verlet integration for greater numerical stability. Pytest testing also encouraged more functional design choices. Packaged with pyproject.toml.

TRAVEL TRACKER

Interactive world map that lets users track countries that they and others have visited. Refactored the project from a multi-page flow to a single-page dialog-driven UI to improve usability and to greatly enhance the visuals. Built with an Express.js backend, PostgreSQL, and with EJS templating.

TO-DO LIST APP

Full-stack to-do app utilising a RESTful API Express.js backend and persistent storage in a PostgreSQL database. Refactored from SSR EJS templating to a CSR architecture, separating frontend and backend and enabling responsive page updates without page reloads. Implemented inline-editing, checkbox-triggered deletion, and robust error feedback and handling.

KEY SKILLS

- Python
- JavaScript / Node.js
- REST APIs
- SQL / PostgreSQL
- NumPy
- Express.js
- VCS / Git / GitHub
- HTML / CSS
- Testing / pytest
- EJS

WORK & EDUCATION

AI/LLM Trainer (RLHF)

Contributed to LLM training via RLHF, evaluating model outputs and providing structured feedback. Specialised over the past 6 months in STEM/mathematics tasks – designing novel problems, probing edge cases in model reasoning, and assessing performance on complex mathematical challenges.

DataAnnotation.tech · 2024–present

Online programming courses

The Complete Full-Stack Web Development Bootcamp – Angela Yu
CS50's Introduction to Programming with Python – David J. Malan

HarvardX & Udemy · 2024–2025

DipHE Psychology

Lancaster University · 2020–2022

DipHE Physics

Durham University · 2016–2018

A-levels

Further Mathematics (A*), Mathematics (A*), Physics (A)

Holy Cross College · 2014–2016